Autonomous Mars Rover: a propositional logic agent

BCA301-5 Artificial Intelligence · AI Express Hackathon · **Group 6**
Track 2: Autonomous Mars Rover (Unit 3, Propositional Logic Agent)

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Repository: `https://github.com/suppprith/mars-rover-logic-agent`

1. PEAS framework

Performance	+1000 docking at the lander with the sample, -1000 losing the rover, -1 per action, -10 for the containment charge. Also logged: path cost, nodes expanded and generated, replans, clauses, resolution steps, reasoning time.
Environment	6x6 grid of safe terrain, unknown hazards and one radiation zone, plus one sample cache. Each square beyond the lander and its neighbours holds a hazard with probability 0.16. Generation retries until the sample is reachable without crossing a hazard. Partially observable, deterministic, sequential, static, discrete, single agent.
Actuators	Drive north, south, east or west to an adjacent square; Collect the sample; Neutralise the radiation zone along a bearing with the single charge; Dock.
Sensors	Hazard warning (hazard adjacent), radiation alert (zone adjacent), sample beacon (cache on this square), neutralise confirmation, bump. Readings are local: the rover never sees a square it has not parked on.

2. Core algorithmic formulation

State space	(position, knowledge base, has_sample, has_charge, radiation_active). The planner searches a subgraph: vertices are squares proved safe or already driven on, less squares proved to hold a hazard; edges join orthogonal neighbours. That vertex set grows with every reading, which is what forces the replan.
Initial state	Rover at the lander (0,0), charge unused, no sample. KB holds Visited(0,0) and the grid adjacency facts. Nothing about hazards, the radiation zone or the cache is given.
Goal test	Mission: Dock at (0,0) while has_sample. Search: n equals the target square, which is the cheapest safe unsurveyed square, or (0,0) once the sample is aboard.
Path cost	g(n) = number of moves from the current square, one unit per edge.
Heuristic	$h(n) = \lvert x_n - x_{goal}\rvert + \lvert y_n - y_{goal}\rvert$, f(n) = g(n) + h(n). Admissible: each move changes Manhattan distance by exactly 1, so no path is shorter than h(n). Consistent: h(n) <= c(n,n') + h(n') = 1 + h(n') for adjacent n, n' , so A* with a closed set is optimal and never reopens a node.
Propositional axioms	$W_{x,y}$ <=> OR over n in Adj(x,y) of H_n and $A_{x,y}$ <=> OR over n in Adj(x,y) of Z_n W hazard warning, A radiation alert, H unknown hazard, Z radiation zone. Added in CNF the first time a square is seen, with unit clauses not H_c, not Z_c for every square driven on and the observed W_c or not W_c.
Entailment	KB entails a query iff KB with the negated query is unsatisfiable. Tested by resolution refutation: set of support seeded on the negated query, a relevance filter dropping clauses over two moves from the query square, maximum clause length 8, and a 2500 step budget. Exhausting the budget returns not proved, never proved false.
First-order rules	Visited(c) => NoHazard(c) · Visited(c) => NoRadiation(c) Visited(c) and NoWarning(c) and Adjacent(c,n) => NoHazard(n) Visited(c) and NoAlert(c) and Adjacent(c,n) => NoRadiation(n) NoHazard(c) and NoRadiation(c) => Safe(c) · RadiationSealed and Cell(c) => NoRadiation(c) Forward chained to a fixpoint, unified against the ground adjacency facts.
Risk, no proof	$P(H_c$ | evidence) = [sum of w(m) over models satisfying the evidence with H_c true] / [sum of w(m) over all models satisfying the evidence], w(m) = product of p over true hazard variables and (1-p) over false, p = 0.16. Enumerated over unproved frontier variables. The rover drives onto the minimum at or below 0.25, otherwise it returns to the lander.

3. Complexity, theory against measurement

Component	Theoretical	Observed, 400 maps
A* per search	$N = n^2$ = 36 squares, branching 4. O(N log N) with a closed set and a consistent heuristic; $O(b^d)$ worst case without one. Space O(N).	3.5 expanded per search, 67.8 per map over 19.3 replans, 117.4 generated
Forward chaining	Adjacency bounded at 4, so $O(n^2)$ ground rule instances. $O(n^2)$ per pass, $O(n^4)$ to fixpoint.	64 facts derived per map, 259 ground facts held
Resolution	$O(2^m)$ worst case for m symbols. $m = 4n^2$ = 144, cut to about 52 by the relevance window, then bounded by the 2500 step budget.	106.1 queries per map, 1298 steps per query against the 2500 cap
Model counting	$O(2^k)$ over k unproved frontier variables, skipped above 2^{16}.	k stays between 2 and 9, so under 512 assignments per call
Whole mission	Turns bounded by the 250 step cap. Memory $O(n^2)$ facts and clauses.	21.1 turns, 19.3 moves, 190.4 clauses, 354.7 ms reasoning; 94.2% survived, 64.5% recovered the sample, mean score 564.2

Measurements: `python run.py --benchmark 400 --seed 0` on Python 3.11. Recorded run: `python run.py --seed 114 --delay 2400`, where resolution proves a hazard at (2,0) and model checking pins the radiation zone at (1,1). 19 turns, 16 moves, 47 nodes expanded, 34 resolution queries, 2 proofs, score 971.